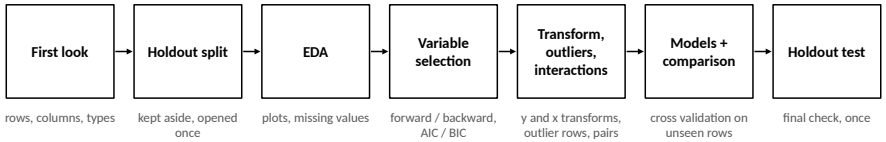
Minjun Cho

Data Analysis

Analysis approach

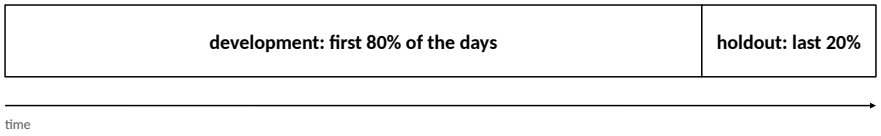


Every choice is made on the development part only. The holdout is opened once, at the end.

Data overview

- **Data:** [tables merged into one: A + B (key = …)]
- **Samples:** [N] rows | **Columns:** [M]
- **Target:** [name] (numeric)
- **Numeric ([n]):** [main examples]
- **Categorical ([n]):** [main examples]
- **Date ([n]):** [all of them]
- **ID ([n]):** [all of them]

Holdout split: time series



- **Split by time:** the last [20]% of the days are the holdout (a day is never cut in two)
- **No shuffling:** the future must not leak into the past
- **development:** [start date] to [end date], [n] rows
- **holdout:** [start date] to [end date], [n] rows
- The holdout is not used until the final test

Holdout split: not time series

development: random 80% of the rows	holdout: random 20%

- **Random split:** 80% of the rows are development, 20% are the holdout (fixed seed, so it can be repeated)
- **development:** [n] rows
- **holdout:** [n] rows
- Rows used to choose variables and models are never used to test them
- The holdout is not used until the final test

EDA: the target

Insert plot here
(histogram of the target)

Insert plot here
(target over time)

- **Left:** [histogram of the target (x = value, y = number of rows)]
- **Right:** [target over time, e.g. the mean by month]
- **Basic statistics:** [mean, median, min, max, sd]
- **Distribution:** [positive? skewed? zeros? range]
- **Over time:** [trend / seasonality / jumps]
- **So what:** [e.g. a log transform may help]

EDA: predictors vs target

Insert plot here
(boxplots by category / scatter plots of the target vs predictors)

- **Left:** [plot of the target vs a predictor (boxplot by category, scatter plot for numeric)]
- **Numeric:** [which move with the target, and in which direction]
- **Categorical:** [which groups differ]
- **Time:** [month / weekday effects]
- **So what:** [candidate predictors]

EDA: relationships between predictors

Insert plot here
(correlation heatmap)

- **Left:** [correlation matrix of the numeric variables (red = positive, blue = negative, white = about 0)]
- **Strongly correlated:** [x and y] ([value])
- **So what:** [keep one / combine / keep both]

EDA: variables with the same information

- **Compared:** [x] and [x_code]: [do they carry the same information?]
- **But:** [one of them has missing values / more levels]
- **Decision:** [keep one / combine / keep both]
  - [numbers that are only labels are treated as categorical variables]

EDA: missing values

Insert plot here
(missing cells / missing count per column)

Insert plot here
(missing pattern / missing rate by group)

- **Left:** [missing cells (red) of each column / missing count per column]
- **Right:** [missing rate by month, or the missing pattern]
- **Missing by column:** [names (% missing)]
- **Pattern:** [missing together? only in some periods?]
- **Rows:** [n] rows with missing values, [n] rows without

EDA: what does the missingness mean?

Insert plot here
(table or plot: mean when missing / present)

- **Left:** [mean of the numeric variables when [x] is missing vs present]
- **Difference:** [big / small], so the missing values [may / may not] have a special meaning
- **How to fill the missing values:** [method, e.g. look back [W] days]
  - numeric: [mean / median of …]
  - categorical: [most frequent value]

Variable selection: AIC and BIC

- **Used linear regression** (lm of the target)
- **Forward selection:** start with no variable and add the one that helps most. **Backward selection:** start with all variables and drop the one that hurts least
- **What is "help"?** Two criteria: AIC and BIC

AIC

AIC = 2k − 2 ln(L)

Pros: Light penalty: keeps more variables, good for prediction

Cons: Can keep noise variables (overfit)

BIC

BIC = k ln(n) − 2 ln(L)

Pros: Heavier penalty (grows with n): a simpler model that is easy to explain

Cons: May drop weak but useful variables (underfit)

k = number of parameters, n = number of rows, L = the maximized likelihood. Smaller is better. When n ≥ 8, BIC charges more for every variable than AIC.

Variable selection: result

Method / measure	Variables	AIC	BIC	Adj. R^2
forward_AIC				
forward_BIC				
backward_AIC				
backward_BIC				
full				

Used [method] since [reason, e.g. AIC is better than BIC when our goal is prediction].
Selected: [target] ~ [x1 + x2 + …]

Assumptions check

Insert plot here
(residuals vs fitted)

Insert plot here
(root of standardized residuals vs fitted)

Insert plot here
(Q-Q plot)

Insert plot here
(residuals vs time)

- **Top-left:** residual vs fitted → [what you see]
- **Top-right:** root(standardized residual) vs fitted → [what you see]
- **Bottom-left:** Q-Q plot → [what you see]
- **Bottom-right:** residual vs time → [what you see]

- **[Target distribution]** → [transformation?]
- **[Variance of residuals vs fitted value]** → [generalized linear model: Poisson (variance ∝ mean), Gamma (variance ∝ mean²)]

Box-Cox Transformations

Insert plot here
(Box-Cox log-likelihood plot)

Insert plot here
(Q-Q plots before / after the transformation)

- **Box-Cox formula:** $y(\lambda) = (y^\lambda - 1) / \lambda$ for $\lambda \neq 0$, $y(\lambda) = \log(y)$ for $\lambda = 0$ (y must be positive; λ is chosen to maximize the log-likelihood of the lm)
- **Left:** [Box-Cox log-likelihood by lambda (dotted lines: 95% interval)]
- **Right:** [Q-Q plots before and after the transformation]
- **Target (Box-Cox):** lambda hat = [value], the nearest simple value is [value], so use [transformation]
- **Predictors:** [transform] of [column] (AIC change [value])

Checkpoint: the model so far

[Model type, e.g. linear regression (lm)] on the [transformed] target

[target] ~ [x1] + [x2] + … (the selected variables + the transformed terms)

In symbols: $\log(\text{TARGET}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \varepsilon$, where X_1 = [first variable]

- **Left:** the model so far ([n] selected variables + [n] transformed terms; no interaction yet)
- **Reading β_1** (the first variable, X_1 = [name]):
  - X_1 increases by 1 unit ([unit]) → log(TARGET) increases by $\hat{\beta}_1$ (other variables fixed)
  - that is, TARGET is multiplied by $e^{\hat{\beta}1}$
  - [if the target is not log-transformed: TARGET itself increases by $\hat{\beta}_1$]
- **Estimate:** $\hat{\beta}_1$ = [__], $\exp(\hat{\beta}_1)$ = [__]

Outliers: how we handled them

Insert plot here
(standardized residuals; red = outlier candidates)

- **Left:** [standardized residuals of the selected lm over time; red dots = outlier candidates]
- **Candidates:** rows with |standardized residual| > [3] (red dots): [n] rows
- **Looked at:** the candidate rows, and how their other columns differ from the normal rows: [what you found]
- **Decision:** [kept / removed / explained by X] because [reason]
- **Also:** MAE is shown next to RMSE, since RMSE reacts strongly to outliers

Interactions

Insert plot here
(interaction plot or test table)

- **Left:** [interaction plot: the target vs [a] for different levels of [b]]
- **Tested:** [pairs] (F-test in the lm)
- **Kept:** [a : b] (p-value [value])
- **Why it makes sense:** [reason]

Candidate models

- **baseline:** the mean of the training y (the number to beat)
- **lm:** ordinary least squares, on the transformed y
- **lasso, ridge:** lm with shrunk coefficients (lasso can drop variables)
- **glm:** Gamma family, log link, on the original y
- **mixed, glmm:** lm / glm plus a random intercept per group

All candidates use the same variables, transformations and interactions.

Measures (1/2): RMSE and MAE

RMSE

RMSE = $\sqrt{(1/n)\,\Sigma\, e^2}$

Pros: Same unit as y; punishes big errors; the standard score

Cons: Sensitive to outliers: a few big errors can dominate

MAE

MAE = $(1/n)\,\Sigma\,|e|$

Pros: Same unit as y; the average miss, easy to explain; robust to outliers

Cons: A big miss counts like several small ones

e = prediction − actual, n = number of rows. All scores are computed on the validation rows, in the original units of y.
Main score for choosing the model: RMSE.

Measures (2/2): MAPE, R^2 and Bias

MAPE

MAPE = (100/n) Σ |e| / |y|

Pros: In percent: comparable across scales

Cons: Explodes when y is near 0 (rows with y = 0 are left out)

R^2

$R^2 = 1 - \Sigma e^2 / \Sigma (y - \text{mean } y)^2$

Pros: No unit: 1 is perfect, 0 is as good as the mean

Cons: Can be negative on new data; hides the size of the error

Bias

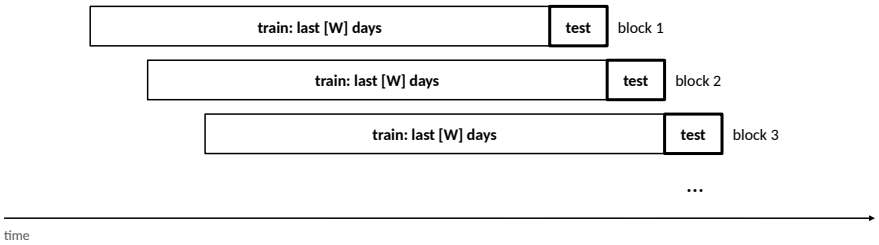
Bias = (1/n) Σ e

Pros: Shows if the model predicts too high or too low

Cons: Errors cancel out: Bias near 0 can hide big errors

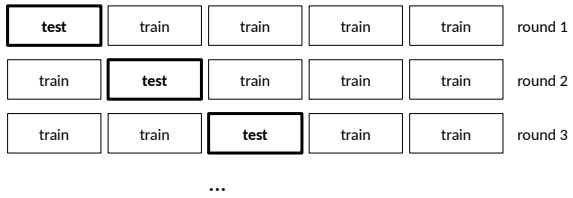
e = prediction − actual, n = number of rows, y = actual.

Cross validation: moving window (time series)



- Train on the last [W] days, predict the next [T] days, then move the window forward by [T] days
- A model only sees days before the days it predicts: no leakage
- All the test predictions are collected and scored together (RMSE)
- Every candidate uses the same blocks, so the scores can be compared

Cross validation: K-fold (not time series)



- The rows are split at random into [5] folds (fixed seed)
- Every fold is predicted by a model trained on the other folds
- All the test predictions are collected and scored together (RMSE)
- Used when there is no time column: random rows can be exchanged

Model comparison

Model	RMSE	MAE	MAPE (%)	R^2	Bias
baseline					
lm					
lasso					
ridge					
glm					
mixed					
glmm					

RMSE bar chart of the models: paste here

- **Top-left:** score table of the 7 models (cross validation, in the original units of y)
- **Bottom-left:** bar chart of the RMSE (smaller is better)
- **Score:** RMSE (main); MAE, MAPE, R^2 and Bias are also shown
- **Best candidate:** [model], RMSE [value]
- **Baseline:** RMSE [value]: the number to beat
- **Others:** [what we see, e.g. lasso is close to lm]

Combinations of steps: ideal vs this analysis

Ideal

- Try many combinations of the steps: variable selection (forward / backward, AIC / BIC), transformations, outlier handling, interactions, models
- Compare all the combinations with cross validation
- Keep the best combination

This analysis

- Limited time: every step was run once, in the order of the previous slides
- The candidate models share the same variables, transformations and interactions
- Only the candidates are compared, with cross validation; the best one is the final model

Final model

[Model]: [family] family, [link] link (fitted on the [original / transformed] y)

[target] ~ [family](mean μ)

g(μ) = [x1 + x2 + … + transformed terms + interactions]

- **Left:** model formula ([model], [link] link)
- **Variables:** [selected] + [transformations] + [interactions]
- **Why [model]:**
  - [reason 1, e.g. matches the variance pattern of the residuals]
  - [reason 2, e.g. the target is positive and skewed]
  - [reason 3, e.g. easy to interpret]
  - [smallest cross-validated RMSE among the candidates, only if true]
- **Cross validation:** RMSE [value] vs baseline [value]

Final model: interpreting a coefficient

[Model], [link] link

$g(\mu) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$

μ = expected [target], X_1 = [first variable]

Interaction in the model: [a × b] (only if X_1 is part of it)

- **Left:** the final model in symbols
- **Reading β_1:**
  - X_1 increases by 1 unit → [g(expected TARGET), e.g. log(expected TARGET)] increases by $\hat{\beta}_1$ (other variables fixed)
  - that is, the expected TARGET is multiplied by $e^{\hat{\beta}_1}$ (log link)
- **Interaction:** [if X_1 is also in a × b, the factor depends on b:]
  - factor = exp($\hat{\beta}_1$ + $\hat{\beta}$_int × [b])
- **Example:** at [b] = [value], the factor is [__]

Final test on the holdout

Holdout score table: paste here

Actual vs predicted: paste here

- **Left:** holdout score table of the models (scores computed on the holdout)
- **Right:** actual vs predicted of the final model on the holdout (points near the diagonal = good)
- **Result:** [model], RMSE [value] on the holdout (baseline [value], cross validation [value])
- **Compared with cross validation:** [similar / worse] because [reason]

Conclusion

Findings

- [main finding 1]
- [main finding 2]
- [main finding 3]

Limitations

- [limitation 1]
- [limitation 2]
- [limitation 3]

Next steps

- [next step 1]
- [next step 2]
- [next step 3]